# ANALYSIS OF COGNITIVE BEHAVIOURAL PATTERNS AMONG STUDENTS USING DATA ANALYTICS

**JESLYN NIVEDHITHA S** **22X020**

**RICHA INFANTA J** **22X037**

DISSERTATION SUBMITTED IN PARTIAL FULFILMENT

OF THE REQUIREMENTS FOR THE DEGREE OF

**BACHELOR OF SCIENCE**

**COMPUTER SYSTEMS AND DESIGN**

OF ANNA UNIVERSITY




**APRIL 2025**

**DEPARTMENT OF APPLIED MATHEMATICS AND COMPUTATIONAL SCIENCES**

**PSG COLLEGE OF TECHNOLOGY**

(Autonomous Institution)

**COIMBATORE-641004**

**PSG COLLEGE OF TECHNOLOGY**

(Autonomous Institution)

**COIMBATORE-641004**

# ANALYSIS OF COGNITIVE BEHAVIOURAL PATTERNS AMONG STUDENTS USING DATA ANALYTICS

BONAFIDE RECORD OF WORK DONE BY


**JESLYN NIVEDHITHA S** **22X020**

**RICHA INFANTA J** **22X037**


DISSERTATION SUBMITTED IN PARTIAL FULFILMENT
OF THE REQUIREMENTS FOR THE DEGREE OF

**BACHELOR OF SCIENCE**
**COMPUTER SYSTEMS AND DESIGN**
OF ANNA UNIVERSITY
**APRIL 2025**

Faculty Guide

Head of the Department

Certified that the candidate was examined in the viva-voce examination held on

Internal Examiner

External Examiner

# TABLE OF CONTENTS


| CHAPTER | | PAGE NO. |
|---|---|---|
| **ACKNOWLEDGEMENT** | | i |
| **SYNOPSIS** | | ii |
| **1. INTRODUCTION** | | 1 |
| 1.1 | BACKGROUND STUDY | 1 |
| 1.2 | COGNITIVE BEHAVIOURAL THERAPY | 1 |
| 1.3 | COGNITIVE DISTORTIONS AND THEIR IMAPCT | 1 |
| 1.4 | PROBLEM STATEMENT | 2 |
| 1.5 | OBJECTIVE | 3 |
| **2. LITERATURE REVIEW** | | 4 |
| 2.1 | MENTAL HEALTH INTERVENTIONS | 4 |
| 2.2 | MACHINE LEARNING IN MENTAL HEALTH PREDICTION | 5 |
| 2.3 | CLUSTERING TECHNIQUES | 7 |
| 2.4 | SUMMARY | 8 |
| **3. METHODOLOGY** | | 9 |
| 3.1 | WORK FLOW | 9 |
| 3.2 | DATA PIPELINE | 9 |
| 3.2.1 | DATA COLLECTION | 10 |
| 3.2.2 | DATA PROCESSING | 10 |
| 3.3 | CLASSIFICATION MODELS | 11 |
| 3.3.1 | LOGISTIC REGRESSION | 11 |
| 3.3.2 | DECISION TREE CLASSIFIER | 11 |
| 3.3.3 | RANDOM FOREST CLASSIFIER | 11 |
| 3.3.4 | XGBOOST CLASSIFIER | 12 |
| **4. EXPERIMENTAL RESULTS AND ANALYSIS** | | 13 |
| 4.1 | DATATSET DESCRIPTION | 13 |
| 4.2 | EVALUATION METRICS | 14 |
| 4.3 | K-MEANS CLUSTERING | 15 |

4.4 VISUALIZATION AND INSIGHTS 19
4.5 CLASSIFICATION MODELS 19
4.6 COMPARITIVE STUDY 25

**CONCLUSION** 28

**REFERENCES** 29

# LIST OF FIGURES


| FIGURE | TITLE | PAGE NO. |
|---|---|---|
| 2.1 | Difference of – CBT vs f-CBT | 4 |
| 2.2 | Comparison of Accuracy | 6 |
| 3.1 | Flow Graph | 9 |
| 4.1 | Age-wise respondents distribution | 13 |
| 4.2 | Gender-wise respondents distribution | 14 |
| 4.3 | Silhouette Score | 16 |
| 4.4 | Pair Plot | 17 |
| 4.5 | Cluster Profiles | 18 |
| 4.6 | Logistic Regression Confusion Matrix | 20 |
| 4.7 | Decision Tree Confusion Matrix | 21 |
| 4.8 | Random Forest Confusion Matrix | 23 |
| 4.9 | XGBoost Confusion Matrix | 24 |
| 4.10 | Accuracy Comparison | 26 |

# LIST OF TABLES


| TABLE | TITLE | PAGE NO. |
|---|---|---|
| 2.1 | Comparison of Models | 5 |
| 2.2 | Comparison of Evaluative Metrics | 7 |
| 4.1 | Result Comparison | 25 |
| 5.1 | Distortion Scores | 27 |

# ACKNOWLEDGEMENT

We express our heartfelt gratitude to **Dr. K. Prakasan**, Principal, PSG College of Technology, for providing us with the necessary facilities and unwavering motivation to carry out this project.

Our profound thanks go to **Dr. Shina Sheen**, Professor and Head, Department of Applied Mathematics and Computational Sciences, PSG College of Technology, for her invaluable moral support and guidance throughout the project.

We extend our deepest appreciation to our Programme Coordinator and guide**, Dr. S. Poomagal**, Associate Professor, Department of Applied Mathematics and Computational Sciences, whose encouragement and support were instrumental in the successful completion of this project.

We are immensely grateful to our tutor, **Dr. M. Sasikumar**, Assistant Professor (Selection Grade), Department of Applied Mathematics and Computational Sciences, for his constant encouragement and kind assistance.

Additionally, we sincerely thank all the staff members of the Department of Applied Mathematics and Computational Sciences for their continuous support and motivation. Finally, we express our heartfelt gratitude to our parents and everyone who contributed to the successful completion of this project.

# SYNOPSIS

A machine learning-based approach is employed to analyse student mental health, focusing on anxiety and depression through classification models. The dataset collected via structured online questionnaires, includes detailed demographic information and responses to Likert-scale items targeting cognitive distortions and mental health symptoms, with a focus on anxiety and depression. The preprocessing pipeline ensures data quality by imputing missing values, encoding categorical variables and scaling features for optimal model performance.

The system employs a suite of classification models to predict mental health clusters, which serve as indicators of students' well-being. These models include Logistic Regression, Decision Tree, Random Forest and XGBoost classifiers. Among these, Logistic Regression achieved the highest accuracy of 98.36%, demonstrating its effectiveness in identifying patterns within the dataset. Each model's performance was evaluated using metrics such as accuracy, classification reports and confusion matrices. Visualizations of these results provide further insights into the models' predictive capabilities.

By leveraging advanced classification techniques, this system identifies key factors influencing student mental health, offering actionable insights for educational institutions and mental health practitioners. These insights aim to facilitate targeted interventions and support strategies, fostering a healthier academic environment. This approach underscores the transformative potential of machine learning in understanding and addressing student mental health challenges.

# CHAPTER 1

# INTRODUCTION

In response to the growing concern over student mental health, this proposed system adopts data-driven methods to explore and address cognitive distortions. By integrating principles of Cognitive Behavioural Therapy (CBT) with machine learning techniques, we aim to provide insights that can inform personalized mental health interventions. The following sections outline the theoretical background, problem statement and objectives that guide this research.

## 1.1 Background Study

Student mental health is a growing concern due to academic pressures and transitions, leading to stress and emotional challenges. Existing support systems are underutilized, highlighting the need for innovative, data-driven methods. This project uses machine learning to analyse cognitive distortions in student data. The goal is to identify patterns of emotional distress and inform personalized mental health interventions through regression models on clustered data. By combining psychological theory and machine learning, this project aims to provide early insights for timely support.

## 1.2 Cognitive Behavioural Therapy

CBT is a structured therapy targeting negative thoughts and behaviours, impacting feelings and actions. It is widely used for anxiety, depression and stress. CBT builds resilience in students, managing academic stress and promoting emotional regulation by restructuring unhealthy thought patterns. A research work by Khizra Khurshid et al., (2024) on CBT reducing procrastination, burnout, test anxiety and self-handicapping in adolescents helps manage unrealistic parental expectations, improving academic performance and school attendance. This aligns with the project's goal of addressing cognitive distortions for better mental health interventions.

## 1.3 Cognitive Distortions and Their Impact

Cognitive distortions are biased, exaggerated or irrational thought patterns that negatively influence one’s perception of reality. These distortions often lead to increased anxiety, low self-esteem and depressive thoughts. In the context of CBT, identifying and challenging these distortions is a crucial step in improving mental health. This specifically focuses on five major cognitive distortions commonly experienced by students:

- **Black-and-White Thinking (All-or-Nothing Thinking):**
  Viewing situations in absolute terms, with no middle ground.
  *Example:* "If I fail this test, I'm a complete failure."
- **Overgeneralization:**
  Drawing broad negative conclusions from a single event.
  *Example:* "I failed once, so I'll always fail."
- **Personalization:**
  Taking responsibility for events beyond one's control or assuming others' actions are related to oneself.
  *Example:* "They didn't invite me, so they must not like me."
- **Catastrophizing:**
  Expecting the worst-case scenario in any given situation.
  *Example:* "If I don't do well on this assignment, I'll ruin my entire future."
- **Rigid 'Should' Statements:**
  Holding unrealistic expectations for oneself or others, leading to guilt or frustration.
  *Example:* "I should never make mistakes, or I'm not good enough."

These distortions are subtle yet powerful contributors to student stress. By identifying them in self-reported data, the project aims to model their impact and predict levels of psychological distress.

## 1.4 Problem Statement

While CBT effectively addresses cognitive distortions, its impact on large student populations is limited by subjective traditional assessments prone to bias and underreporting. High levels of cognitive distortion may go unnoticed. A lack of data-driven approaches to quantitatively identify cognitive patterns creates a gap between mental health resources and student needs. This leads to delayed intervention and worsening conditions. Studies in Pakistan show rising academic stress and emotional disturbance, highlighting the need for intervention-based research. The research paper on reducing academic burnouts and self-handicapping indicates CBT improves procrastination and test anxiety among adolescents and helps them manage parental expectations. This project addresses the gap by applying machine learning to student mental health data, modelling cognitive distortions to complement existing psychological support systems and enable scalable analytics.

## 1.5 Objective

The primary objectives of this project are:

- To explore the role of cognitive distortions in student mental health using CBT principles.
- To identify and analyse five specific cognitive distortions that affect students' thinking patterns.
- To preprocess and cluster student mental health data for effective modelling.
- To train and evaluate regression models that can predict the severity or presence of distortions based on input features.
- To visualize residuals and feature importances to gain deeper insight into the models' decision-making.
- To demonstrate the feasibility of integrating psychology and data science in student wellness initiatives.

# CHAPTER 2

# LITERATURE REVIEW

Mental health interventions are increasingly enhanced by data-driven methods with Cognitive Behavioural Therapy (CBT) as a key framework for addressing cognitive distortions. Machine learning enables earlier, more accurate detection of conditions like anxiety and depression. These techniques help uncover patterns in mental health data, supporting personalized strategies for student well-being.

## 2.1 Mental Health Interventions

Mental health interventions have increasingly become a critical focus within academic institutions. Students, particularly those in universities, face escalating levels of psychological stress due to academic expectations, social dynamics and personal identity development. Cognitive Behavioural Therapy (CBT) has emerged as a widely accepted psychological framework to help individuals restructure negative thought patterns, including maladaptive beliefs and irrational thinking. You et al., (2022) compared face-to-face CBT (f-CBT) and internet-delivered CBT (i-CBT) interventions and found both methods effective for reducing anxiety symptoms in university students. The i-CBT approach, in particular, was praised for its scalability, cost-effectiveness and ability to reduce stigma related to therapy-seeking. Figure 2.1 shows the difference between the i-CBT and f-CBT observed.

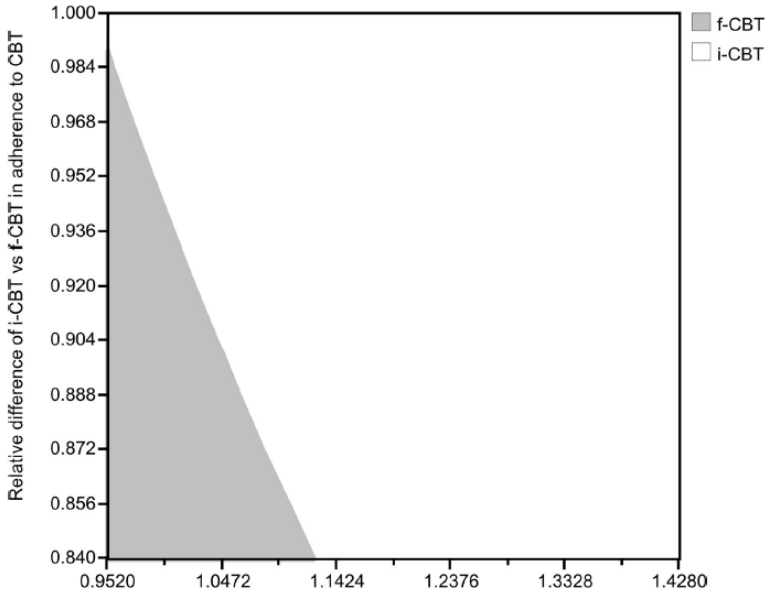



**Figure 2.1 Difference of i-CBT vs f-CBT**

Figure 2.1 represents the relative difference in adherence, with f-CBT showing higher adherence compared to i-CBT within the shaded range. The x-axis shows a range of adherence metrics, while the y-axis indicates the relative difference of i-CBT vs. f-CBT. The area outside the shaded region favors i-CBT, while the shaded region indicates superiority of f-CBT in adherence.

Khurshid et al., (2025) further demonstrated that CBT not only alleviated anxiety and academic burnout but also helped students cope with social stressors and unrealistic parental expectations. Their randomized control trial showed improvements in school attendance and emotional well-being, reinforcing CBT's role in fostering student resilience. Oud et al., (2019) expanded the scope of CBT research by focusing on its efficacy for children and adolescents with depression. Their systematic review and meta-regression analysis of 31 randomized controlled trials revealed that CBT is moderately effective in reducing depressive symptoms at the end of treatment and at follow-up. Notably, the inclusion of specific components behavioural activation and challenging thoughts alongside caregiver involvement significantly enhanced long-term outcomes.

## 2.2 Machine Learning in Mental Health Prediction

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have enabled researchers to analyse psychological data with unprecedented depth and scale. Jiang et al (2024) introduced a framework to extract cognitive pathways from social media text using Deep Learning and Large Language Models (LLMs). They annotated user-generated content with CBT-based categories and used models like ERNIE 3.0 and GPT-4 to classify and summarize thought patterns.

**Table 2.1 Comparison of the models**

| Models | Level | Precision | Recall | F1 |
|---|---|---|---|---|
| ERNIE 3.0 | Parent Nodes | **74.92** | **77.29** | **76.64** |
| | Child Nodes | 40.21 | 41.33 | 46.70 |
| | Overall | 66.30 | 58.83 | 62.34 |
| GPT -4 | Parent Nodes | 31.69 | 41.47 | 35.93 |
| | Child Nodes | 15.44 | 19.45 | 17.22 |
| | Overall | 23.43 | 30.07 | 26.34 |
| GPT-3.5 | Parent Nodes | 28.61 | 36.50 | 32.08 |
| | Child Nodes | 12.19 | 14.92 | 13.42 |
| | Overall | 20.27 | 25.31 | 22.51 |

Table 2.1 shows the evaluative measures across each model. Their study shows that supervised learning models (like ERNIE 3.0) performed better in identifying structured CBT components (ABCD: Activating Event, Belief, Consequence, Disputation), while GPT-4 excelled in summarization tasks. This reinforces the value of combining clinical theories with AI models for scalable psychological assessment.

Vuyyuru et al., (2023) expanded on these ML applications by developing a Transformer-CNN hybrid model specifically designed for psychological assessment and intervention in CBT. Their novel approach combined the contextual understanding capabilities of Transformer architecture with the feature extraction power of Convolutional Neural Networks (CNNs). The hybrid model demonstrated an impressive 97% accuracy in identifying depression from user-generated content, substantially outperforming traditional methods and standalone models. This research highlights the potential of hybrid neural architectures to significantly enhance diagnostic accuracy in mental health applications, particularly when processing complex clinical text data such as patient narratives and therapy transcripts.

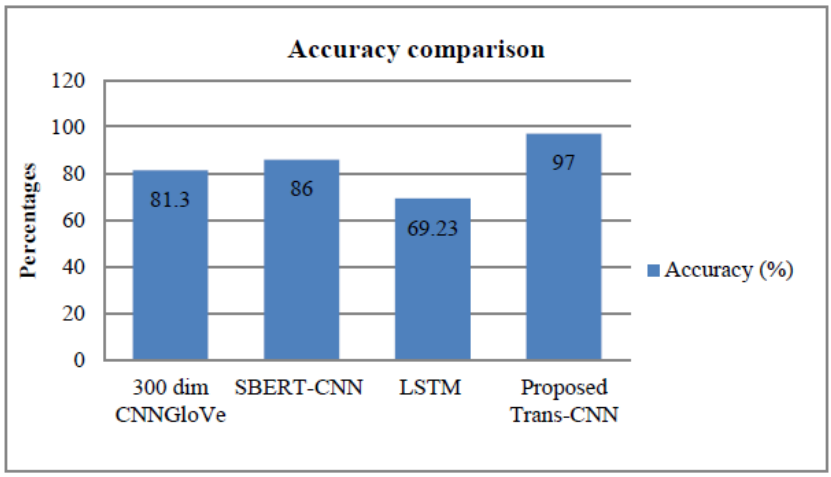



**Figure 2.2 Comparison of accuracy**

Figure 2.2 shows the accuracy comparison across the different models. Table 2.2 compares the measures that provides better analysis of the working of each model. Furthermore, integrating ML into mental health not only enables early detection of disorders such as depression or anxiety but also improves personalization and reduces reliance on limited clinical resources. The rise of chatbots, digital therapists, and AI-powered screening tools is transforming how CBT is administered in real time.

**Table 2.2 Comparison of the evaluative metrics**

| Method | Precision (%) | Recall (%) | F1-score (%) |
|---|---|---|---|
| CNN | 72 | 87 | 79 |
| SBERT-CNN | 85 | 87 | 86 |
| LSTM | 74 | 79 | 77 |
| **Proposed Trans-CNN** | 91 | 95 | 94 |

Table 2.2 compares the performance of various models proposed by Chen et al., (2023) based on precision, recall, and F1-score. The proposed Trans-CNN model outperforms all others, achieving the highest scores across all metrics. This indicates its superior effectiveness in accurately identifying mental health patterns.

## 2.3 Clustering Techniques

Data segmentation and predictive modelling are vital in understanding mental health dynamics. Clustering helps in identifying patterns or profiles among students experiencing similar cognitive distortions, while regression models can predict the likelihood or severity of emotional distress based on features such as responses, behaviours or demographics. Jiang et al., (2024) used hierarchical classification to segment user input into CBT-related categories, while Vuyyuru et al., (2023) applied clustering principles more directly through their Trans-CNN model. This model leveraged Transformer encoders with self-attention mechanisms to capture contextual dependencies in clinical text. The self-attention process generated query, key and value vectors for each token, enabling the model to focus on meaningful semantic relationships—akin to clustering techniques that identify patterns in data. Oud et al., (2019) further contributed by using meta-regression to identify factors influencing CBT effectiveness for youth depression. Their findings showed that combining behavioural activation with cognitive restructuring led to improved outcomes and highlighted caregiver involvement as a structural factor enhancing treatment efficacy. These insights suggest that future clustering approaches could benefit from considering contextual elements such as family dynamics and intervention settings.

Vuyyuru et al., (2023) research also introduced sophisticated concatenation methods to fuse features from both Transformer and CNN architectures before classification, creating a unified representation that leverages the complementary strengths of both approaches. Their experimental results on a large depression dataset demonstrated that hybrid models offer superior performance in both

classification accuracy (97%) and balanced metrics compared to standalone approaches like XGBoost or LSTM models.

## 2.4 Summary

The literature strongly supports the effectiveness of CBT in educational settings and reveals promising opportunities for using AI and machine learning to enhance its application. However, current research reveals the following gaps:

- i-CBT appears more cost-effective than f-CBT for university students with mild anxiety in Hong Kong, yielding higher QALYs at a lower cost. However, the cost-effectiveness of i-CBT is highly dependent on patient acceptance and adherence to the internet platform. i-CBT could improve the efficiency of health service delivery and maximize favourable treatment outcomes.
- This randomized controlled trial found that CBT effectively reduced procrastination, academic burnout, test anxiety and self-handicapping behaviours among adolescents facing academic problems. CBT also significantly improved students' ability to manage unrealistic parental expectations and enhanced their school functioning, including attendance and punctuality. The findings suggest CBT is a useful intervention for addressing emotional and academic challenges in adolescents.
- AI-Enhanced CBT leverages deep learning to classify cognitive pathways from social media texts, achieving a micro-F1 score of 62.34%. GPT-4 excels in summarizing these pathways (Rouge-1 score of 54.92) but has hallucination issues. The study introduces a hierarchical text classification task with four main categories (ABCD) and nineteen subcategories, along with a text summarization task.
- This psychological assessment introduces a Transformer-CNN hybrid model for Cognitive Behavioural Therapy (CBT) in psychological assessment and intervention, aiming to enhance diagnostic accuracy and treatment efficiency. The model combines the Transformer's ability to capture contextual dependencies with the CNN's strength in extracting local and global features from clinical text data. Experiments showed the hybrid model achieved an accuracy of 97%, outperforming traditional CBT methods and standalone models.

# CHAPTER 3

# METHODOLOGY

Understanding patterns in student mental health requires the application of advanced machine learning techniques to identify predictors of anxiety and depression. The methodology incorporates data preprocessing and model training to generate meaningful insights. By leveraging classification models, it predicts mental health indicators based on survey responses and demographic features.

## 3.1 Workflow

The system is structured around a modular architecture, designed to facilitate efficient processing and analysis of student psychological data. It consists of interconnected components, each serving a distinct purpose in the data processing workflow. The architecture begins with a Data Collection Interface, followed by a Data Preprocessing Module that ensures data quality and consistency. Next, a Feature Engineering and Selection Module enhances the dataset by deriving relevant features. The core analytical component is the Model Training and Regression Engine, which leverages machine learning algorithms to predict mental health indicators. This modular design supports scalability, maintainability, and flexibility, ensuring the system can adapt to evolving needs in understanding cognitive and emotional patterns among students. Figure 3.1 shows the flow of the system methodology.

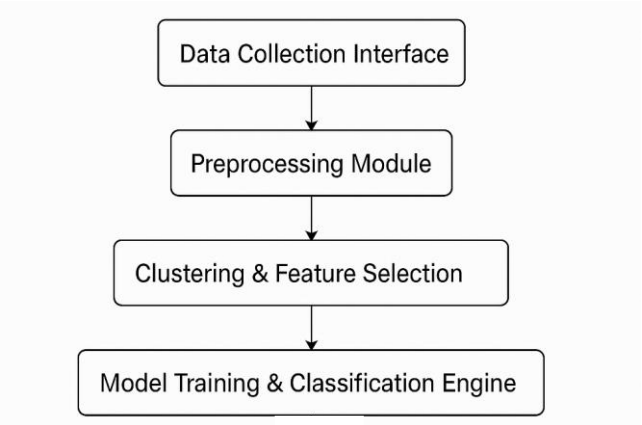



**Figure 3.1 Flow Graph**

## 3.2 Data Pipeline

The data pipeline serves as the backbone of the analytical workflow, managing the journey of data from initial input to actionable insights. It ensures consistency, quality, and structure throughout the

process. Each stage of the pipeline is designed to transform raw survey responses into meaningful information. The following subsections describe the major components of the pipeline in detail.

### 3.2.1 Data Collection

Data is collected through structured online questionnaires created using Google Forms. The questionnaire includes:

- **Demographic Information**: Age, gender, academic year, course details.
- **Likert-Scale Responses**: Questions targeting cognitive distortions (e.g., catastrophizing, overgeneralization) and mental health symptoms (e.g., anxiety, depression). The collected data is stored in CSV format (original_student_gform.csv) for further processing.

### 3.2.2 Data Preprocessing

Various Python libraries were used to implement data preprocessing, clustering and classification tasks efficiently. These tools provided functionality for data handling, modelling and visualization. The key libraries and their roles in the project are as follows:

- **numpy**: Used for handling numerical operations and missing value imputation.
- **re (regex) and string**: Applied for extracting numerical data from text and cleaning string fields like age ranges or course names.
- **matplotlib and seaborn**: For generating a variety of visualizations such as heatmaps, bar plot and cluster profiles.
- **scikit-learn**: Core machine learning toolkit used for:
  - Imputation (`SimpleImputer`)
  - Feature scaling (`StandardScaler`)
  - Train-test splitting
  - KMeans clustering
  - Model evaluation metrics (accuracy, confusion matrix, classification report)
- **xgboos**t: Used for building the XGBoost classifier, which achieved the highest predictive performance in this study.

## 3.3 Classification Models

The system utilizes a set of supervised learning algorithms to classify students into mental health risk clusters based on their CBT distortion patterns, anxiety and depression scores. Each model brings unique capabilities in pattern recognition, interpretability and generalization performance. The models were trained using a pre-processed dataset with labelled cluster information obtained from K-Means clustering algorithm.

### 3.3.1 Logistic Regression

Logistic Regression is a linear model that estimates the probability of a data point belonging to a particular class using a logistic (sigmoid) function. It assumes a linear relationship between the input features and the log-odds of the target class, making it well-suited for linearly separable data. The model computes coefficients for each feature, providing insight into how strongly a feature influences classification. This makes it especially valuable for feature interpretation. Logistic Regression was used as a baseline model and its simplicity allows for fast training and good generalization for low-dimensional data.

### 3.3.2 Decision Tree Classifier

A Decision Tree classifier splits the dataset into subsets based on feature values using criteria such as Gini impurity or entropy (information gain). The model builds a tree structure where each internal node represents a decision on a feature and each leaf node corresponds to a class label. It is capable of capturing non-linear relationships and interaction effects between features. The tree structure is inherently interpretable, as the prediction path for any input can be visually traced. Decision Trees handle both numerical and categorical data and are robust to irrelevant features, making them suitable for complex mental health survey data.

### 3.3.3 Random Forest Classifier

Random Forest is an ensemble model that builds multiple Decision Trees on different random subsets of the training data and averages their outputs for classification. It introduces randomness by selecting a subset of features for each split, which reduces correlation between trees and improves generalization. The model reduces overfitting compared to a single Decision Tree and improves accuracy and robustness through bagging (bootstrap aggregation). Random Forests provide feature

importance scores, helping to identify which psychological or behavioural attributes most influence the clustering. The ensemble nature makes it more stable against noise and data variance.

### 3.3.4 XGBoost Classifier

XGBoost (Extreme Gradient Boosting) is a powerful boosting algorithm that builds models sequentially, where each new model attempts to correct the errors of its predecessor. It minimizes a custom loss function using gradient descent and integrates regularization techniques (L1 and L2) to control model complexity. XGBoost is known for its high efficiency, parallel processing and sparse data handling, making it suitable for large and complex datasets. The model is particularly effective in capturing complex, non-linear interactions between features and class labels. Like Random Forest, XGBoost also provides feature importance rankings, enabling deeper insight into the most influential psychological indicators of mental health.

# CHAPTER 4

# EXPERIMENTAL RESULTS AND ANALYSIS

A structured approach was followed to design the experimental setup and analyse outcomes from the student mental health dataset. It covers data preprocessing, K-Means clustering, classification model implementation and a comparative study of model performances. The aim is to provide a comprehensive overview of the methodologies applied and the insights gained at each stage of the analysis.

## 4.1 Dataset Description

The dataset comprised 303 responses collected from a survey administered to students. This survey aimed to capture various aspects of their mental health, cognitive patterns, and demographic information. The dataset included both numerical and categorical features, providing a comprehensive view of the factors influencing student well-being.

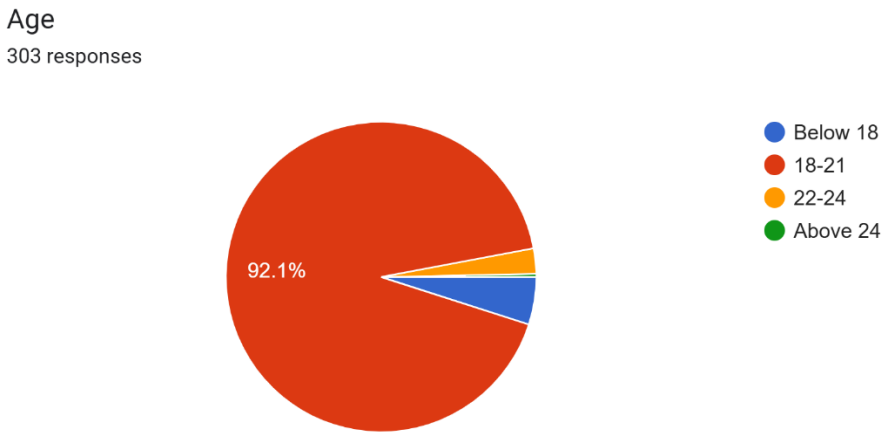



**Figure 4.1 Gender-wise respondents distribution**

Figure 4.1 displays the age distribution of 303 student respondents. A dominant 92.1% of participants are between 18 and 21 years old, suggesting the sample mainly consists of undergraduate students. Smaller age groups include those Below 18, 22–24 and Above 24. The minimal representation from older age brackets indicates a youthful participant pool.

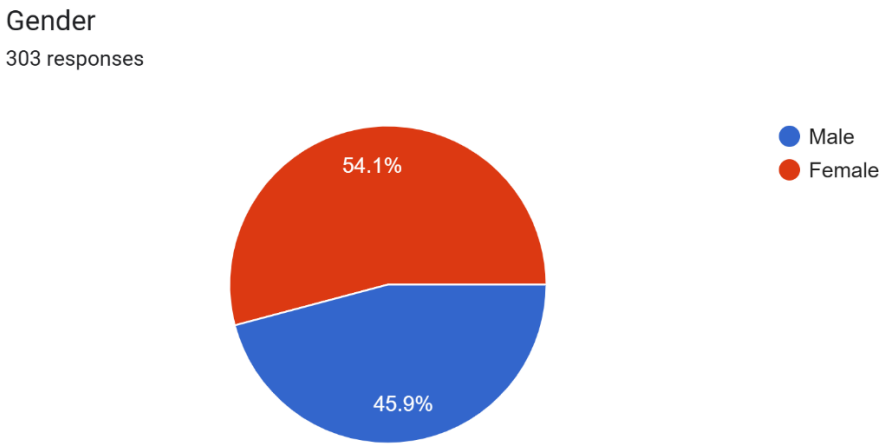



**Figure 4.2 Age-wise respondents distribution**

Figure 4.2 represents the gender distribution among 303 student respondents. Female participants make up the majority at 54.1%, while males account for 45.9% of the total. The distribution is relatively balanced, with a slight female majority. This gender spread provides a fair basis for gender-related mental health analysis.

## 4.2 Evaluation Metrics

To assess the predictive performance of each classification model in accurately assigning mental health clusters, the system utilizes a combination of core evaluation metrics. These metrics help understand not only overall model accuracy but also the model's ability to handle imbalanced clusters and multi-class classification challenges.

1. **Confusion Matrix**

- A square matrix showing actual vs. predicted cluster labels.
- Helps visualize how often a model confuses one mental health cluster for another.
- Heatmaps were generated for each model to intuitively interpret cluster-level prediction strengths and weaknesses.

2. **Accuracy**

Accuracy represents the overall correctness of the model's predictions, indicating how many instances were correctly classified out of the total. However, accuracy can be misleading if the classes are imbalanced. It is calculated in the Equation (4.1).

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (4.1)$$

where, TP is True Positives, TN is True Negatives, FP is False Positives, and FN is False Negatives.

3. **Precision**

Precision measures the proportion of correctly predicted positive instances out of all instances predicted as positive. It reflects how well the model minimizes false positives.
It is calculated in Equation (4.2).

$$\text{Precision} = \frac{TP}{TP+FP} \quad (4.2)$$

4. **Recall**

Recall indicates the model's ability to correctly identify all actual positive instances. It shows how effectively the model captures true positives while minimizing false negatives. It is calculated in Equation (4.3).

$$\text{Recall} = \frac{TP}{TP+FN} \quad (4.3)$$

5. **F1-Score**

The F1-score is the harmonic mean of precision and recall, providing a balanced measure when both false positives and false negatives are important. It is especially useful in cases of class imbalance. It is calculated in Equation (4.4).

$$\text{F1-score} = \frac{2\times(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})} \quad (4.4)$$

## 4.3 K-Means Clustering

K-Means clustering was employed to identify distinct groups of students based on their mental health characteristics. The primary goal was to segment the student population into clusters with similar cognitive and emotional profiles.

**Feature Selection**

Key features related to anxiety, depression, and cognitive distortions were selected for clustering. Specifically, the features used were "Black & White Thinking", "Anxiety", and "Depression," along with "Overgeneralization", "Personalization", "Catastrophizing", and "Should Statements," as defined in the `cluster_features` list.

**Data Scaling**

To ensure that all features contributed equally, the data was scaled using StandardScaler **,** which transforms each feature to have a mean of 0 and a standard deviation of 1. This is done by subtracting the mean of the feature from each data point and dividing by the standard deviation. Scaling is essential because it prevents features with larger scales from dominating the model and improves convergence for algorithms sensitive to feature magnitude, such as gradient descent-based optimization.

**Optimal Clusters**

The silhouette scores were computed for a range of cluster numbers (e.g., 2 to 10), and a silhouette plot was generated to visualize the clustering quality for each number of clusters. To calculate the silhouette score, for each point, calculate the mean distance to all other points in the same cluster (a) and the mean distance to all points in the next nearest cluster (b). Then, the Silhouette Coefficient for that point is calculated in Equation (4.5).

$$\text{Silhouette Coefficient} = \frac{(b-a)}{\max(a,b)} \tag{4.5}$$

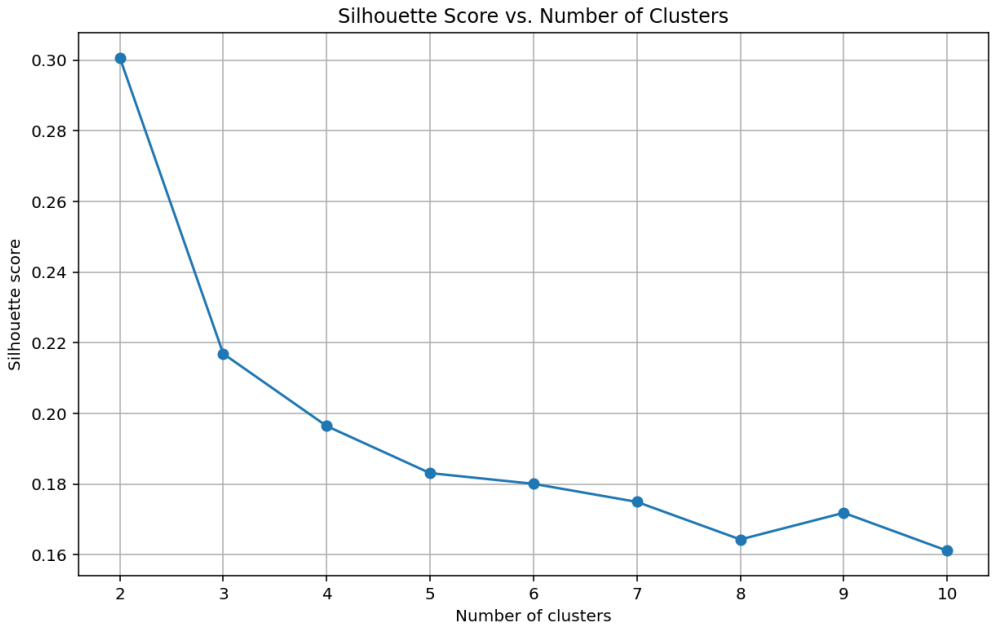



**Figure 4.3 Silhouette score**

Figure 4.3 shows the Silhouette Score for different numbers of clusters. The Silhouette Score measures how similar each data point is to its own cluster compared to other clusters. It ranges from -1 to 1, where a higher score means better-defined clusters. In this plot, the highest score is approximately 0.3 at 2 clusters, so we chose K=2 for K-Means because that leads to the most meaningful clusters for this particular dataset. This indicates that, on average, data points are reasonably well-clustered.

**Cluster Visualization**

Scatter plots were generated to visualize the clusters in reduced dimensions (e.g., using PCA or t-SNE). These plots showed how students were grouped based on the selected features.

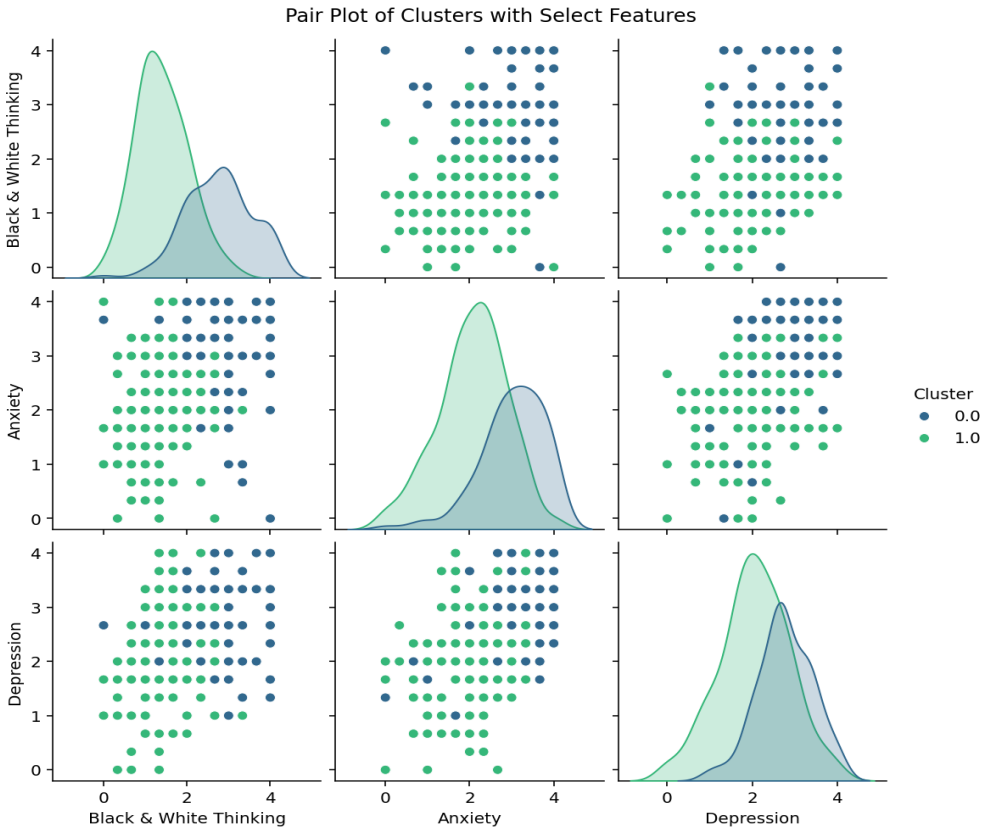



**Figure 4.4 Pair Plot**

Figure 4.4 displays relationships between pairs of variables ("Black & White Thinking", "Anxiety", "Depression") with each point coloured by its cluster assignment (0 or 1). In this pair plot, there are multiple scatter plots as well as distribution plots. The scatter plots display how all the points relate to each other according to various categories. Points coloured with the same hue show that the students are in the same cluster. If the points are located close to each other on the plot, it means the values are similar. This helps visualize how well the clusters are separated based on these variables, for

instance, if the points of different colours are well-separated in a scatter plot, it indicates that the clusters are distinct for those variables. For example, we can see a general trend that students with a "Cluster" of 0 (the blue hue) tend to score higher on anxiety when compared to the students with a cluster of 1 (the green hue). The distribution plots, which are diagonal on the image, show the distribution of the data for each column. We can see the cluster has one normalized curve for each cluster for example, the Black and White Thinking column has two curves that show how the data for that variable is distributed among the two clusters.

**Cluster Profile Heatmaps**

Heatmaps were created to display the average levels of anxiety, depression and CBT distortions within each cluster. These heatmaps provided insights into the characteristics of each cluster.

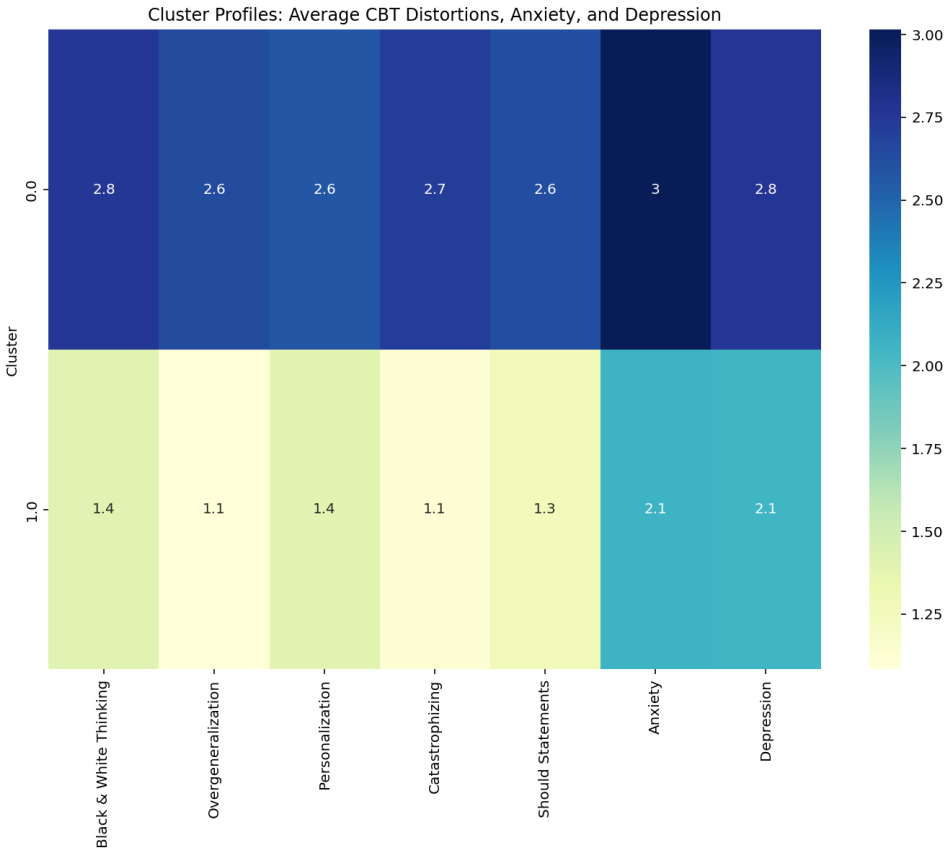



**Figure 4.5 Cluster Profiles**

Figure 4.5 shows the average levels of different features (cognitive distortions, anxiety, and depression) for each cluster. Each row represents a cluster (0 and 1), and the colour intensity indicates the average level of each feature within that cluster. The numerical values are also overlaid on each of

the box, to help you analyse the data. Looking at the data we can determine that Cluster 0 has much higher levels of all cognitive distortions, anxiety and depression compared to Cluster 1. For example, the mean of "Black & White Thinking" for students in cluster 0 is 2.8 while cluster 1 is only 1.4. The implementation identified 2 clusters based on the analysis of silhouette scores. The average silhouette score was 0.3, indicating a reasonable cluster separation.

## 4.4 Visualization And Insights

A variety of visual tools were used to explore the results of the classification models, understand cluster behaviour, and draw data-driven insights for intervention strategies.

1. **Confusion Matrix Heatmaps:**

    - Each classifier generated a separate heatmap for its confusion matrix.
    - These helped assess how consistently models predicted each mental health group, and where misclassifications occurred.

2. **Accuracy Comparison Bar Chart:**

    - A bar plot compared the accuracy scores of Logistic Regression, Decision Tree, Random Forest, and XGBoost.
    - This visual made it clear that XGBoost significantly outperformed all other models, serving as the most reliable option for deployment.

3. **Cluster Profile Heatmaps**

    - Heatmaps were created to show the average CBT distortion, anxiety, and depression levels across each identified cluster.
    - This revealed which clusters were high-risk and what kind of distortions (e.g., Catastrophizing, Overgeneralization) were most prevalent.

## 4.5 Classification Models

Four classification models were trained to predict the cluster membership of students: Logistic Regression, Decision Tree, Random Forest, and XGBoost. The objective was to develop models that

could accurately classify students into their respective mental health clusters based on their survey responses. The dataset was split into training (80%) and testing (20%) sets. Splitting the data ensures the model is evaluated on unseen data, providing a realistic assessment of its performance. Each model was trained using the training set, with hyperparameter tuning to optimize performance. Training the model involves learning the patterns in the data, while hyperparameter tuning optimizes the model's settings to achieve the best possible performance.

### Logistic Regression Result

The Logistic Regression model achieved an accuracy of 98.36%. The confusion matrix reveals its impressive classification capability. It correctly predicted the majority of instances in both clusters with very few misclassifications. The model achieved high true positive and true negative rates, ensuring reliable performance. Only one false negative was observed, highlighting a minor limitation in Cluster 0 prediction. Overall, the matrix confirms the model's robustness and precision in binary classification.

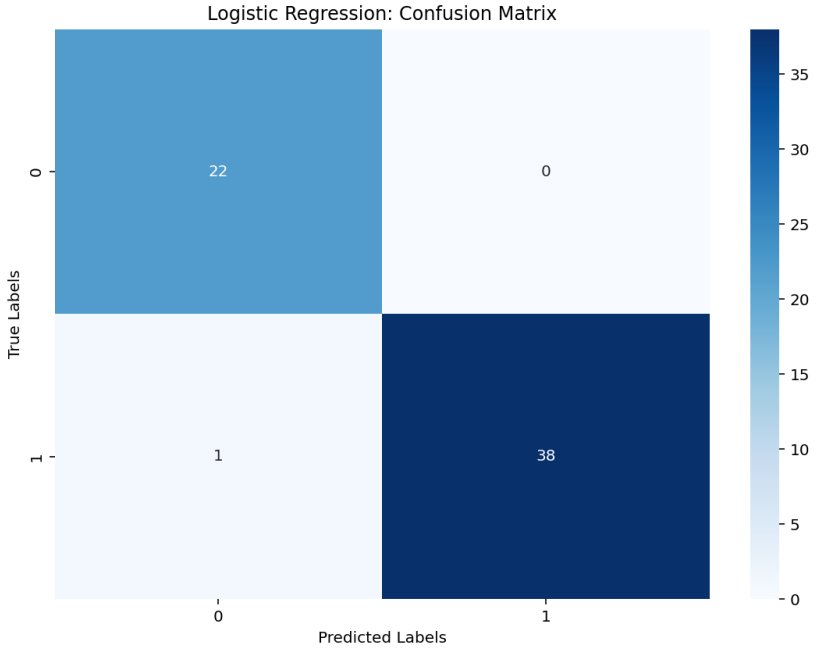



**Figure 4.6 Logistic Regression Confusion Matrix**

Figure 4.6 shows the performance of the Logistic Regression model in classifying two clusters. It correctly classified 22 instances of Cluster 0 and 38 instances of Cluster 1. There were no false positives for Cluster 0, and only 1 false negative for Cluster 1. This results in a high classification accuracy with very minimal misclassification. The model demonstrates strong predictive ability, especially in correctly identifying true positives.

The classification report for Logistic Regression is as follows:

**Cluster 0:** Precision = 100%, Recall = 95.65%, F1-score = 97.78%

**Cluster 1**: Precision = 97.44%, Recall = 100%, F1-score = 98.70%

Inference

Logistic Regression performed exceptionally well in classifying both clusters, especially Cluster 1 (100% recall), which indicates reliability in correctly identifying all actual negatives. The single false negative in Cluster 0 suggests a very minor weakness in detecting all positives. Overall, it's a robust and interpretable model suitable for binary classification when simplicity and high performance are both desired.

**Decision Tree Result**

The Decision Tree model achieved an accuracy of 91.80%**.** The confusion matrix provides a clear overview of its classification performance. It illustrates how well the model distinguishes between the two clusters based on true and predicted labels. This matrix helps identify areas where the model excels and where it misclassifies data. By analysing true positives, false positives, false negatives, and true negatives, deeper insights are drawn. It serves as a valuable tool to assess the model's strengths and limitations in binary classification.

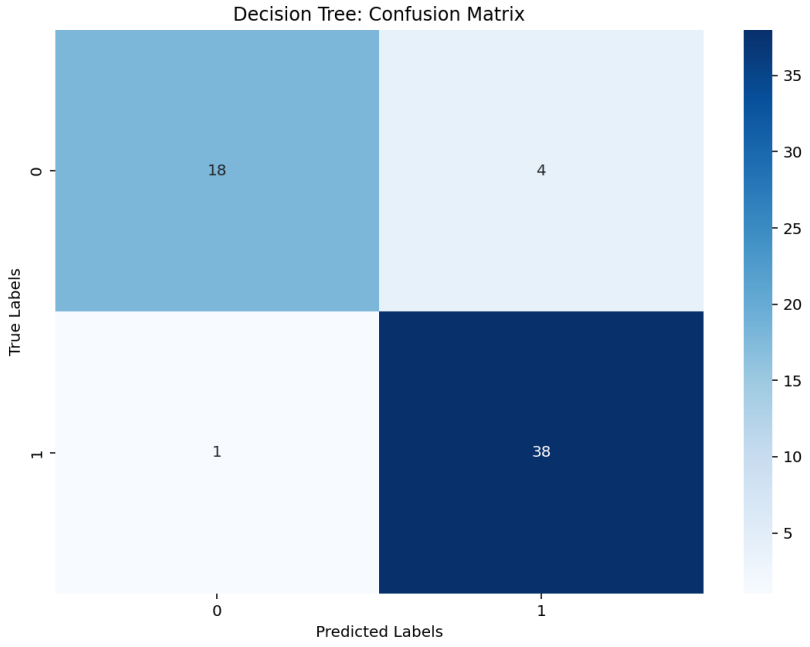



**Figure 4.7 Decision Tree Confusion Matrix**

Figure 4.7 highlights the decent performance in binary classification for the Decision Tree model. It accurately predicted 18 samples of Cluster 0 and 38 samples of Cluster 1. However, 4 instances from Cluster 0 were misclassified as Cluster 1, showing a drop in recall. Cluster 1 was handled more effectively, with only 1 misclassification. Overall, the matrix indicates the model's bias towards predicting Cluster 1 and a slight weakness in identifying all positives from Cluster 0.

**Cluster 0**: Precision = 95%, Recall = 81.82%, F1-score = 88%

**Cluster 1**: Precision = 90.48%, Recall = 97.44%, F1-score = 93.75%

Inference

The Decision Tree model performed reasonably well but struggled slightly in predicting Cluster 0, as seen from the 4 false negatives. This indicates the model tends to misclassify Cluster 0 instances as Cluster 1. Although precision is high, recall drop for Cluster 0 highlights the model's limited ability to detect all positive samples for that class.

**Random Forest Result**

The Random Forest model achieved an accuracy of 98.36%, same as Logistic Regression highlighting its exceptional classification performance. It provides a visual representation of the model's predictions compared to actual class labels. With only one misclassification, the matrix confirms the model's high accuracy and reliability. The perfect recall for Cluster 1 demonstrates its strength in identifying all positive instances. This performance reflects the robustness and generalization power of ensemble learning in Random Forest.

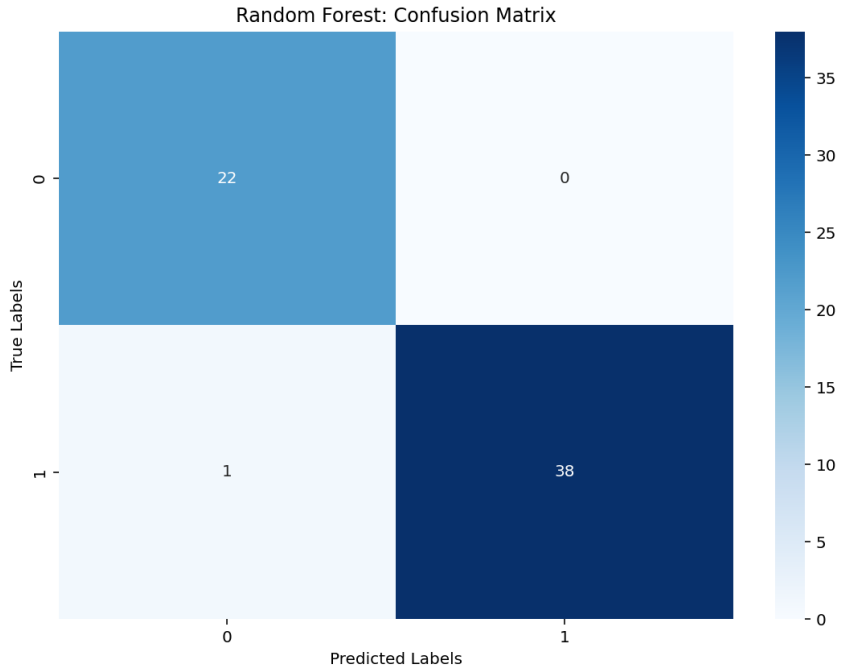



**Figure 4.8 Random Forest Confusion Matrix**

Figure 4.8 shows the Random Forest model accurately classified 22 instances of Cluster 0 and 38 of Cluster 1. Only one instance from Cluster 0 was misclassified, with no false negatives for Cluster 1. This resulted in an overall accuracy of 98.36%, matching Logistic Regression. Cluster 1 achieved 100% recall, indicating perfect identification of all its samples. The figure reflects the model's strong predictive power and reliability due to ensemble learning.

**Cluster 0**: Precision = 100%, Recall = 95.65%, F1-score = 97.78%

**Cluster 1**: Precision = 97.44%, Recall = 100%, F1-score = 98.70%

Inference

Random Forest delivered highly accurate results, with only a single misclassification. Its ability to correctly classify all instances of Cluster 1 (100% recall) and most of Cluster 0 suggests strong generalization and robustness, potentially due to ensemble learning.

**XGBoost Result**

The XGBoost model achieved perfect accuracy of 100%. The confusion matrix illustrates flawless classification performance. It presents a perfect match between predicted and actual labels, with zero misclassifications. Such results demonstrate the model's exceptional accuracy and predictive strength.

The matrix reflects the effectiveness of gradient boosting in capturing complex patterns. However, further validation is recommended to ensure the model's performance is not due to overfitting.

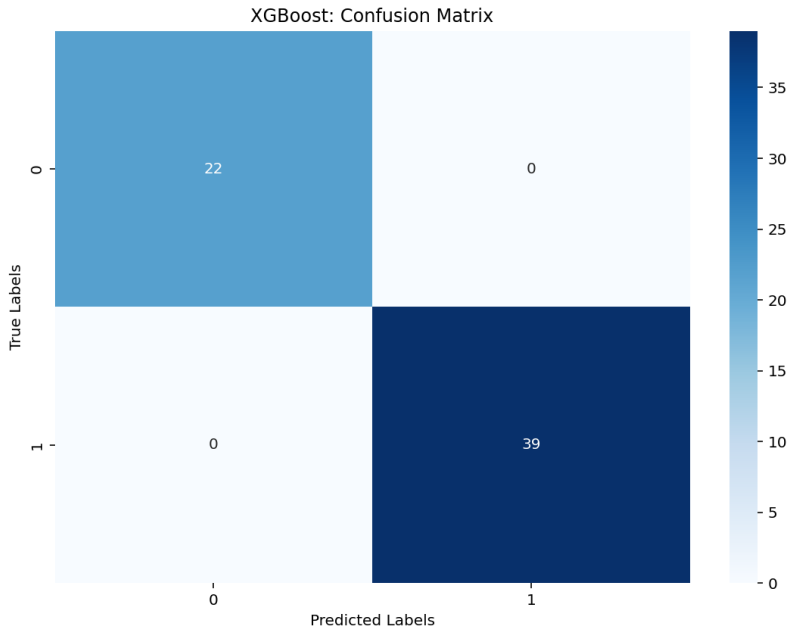



**Figure 4.9 XGBoost Confusion Matrix**

Figure 4.9 shows perfect classification performance with 100% accuracy using XGBoost. Both the true positive and true negative counts are maximized, indicating that all instances in the dataset are correctly classified. The matrix demonstrates that there are no misclassifications, with the model achieving optimal precision and recall for each class.

**Cluster 0:** Precision = 100%, Recall = 100%, F1-score = 100%

**Cluster 1:** Precision = 100%, Recall = 100%, F1-score = 100%

Inference

XGBoost outperformed all other models, with zero misclassifications. This indicates exceptional predictive power and model optimization, likely owing to its gradient boosting framework. However, such perfection may also hint at potential overfitting—cross-validation or testing on unseen data is advised to confirm generalizability.

## 4.6 Comparative Study

This section presents a comparative analysis of the four machine learning models applied to the classification task—**Logistic Regression**, **Decision Tree**, **Random Forest**, and **XGBoost**—based on their accuracy, precision, recall, and F1-scores.

**Table 4.1 Result Comparison**

| **Model** | **Accuracy** | **Precision (C0/C1)** | **Recall (C0/C1)** | **F1-Score (C0/C1)** | **Misclassifications** |
|---|---|---|---|---|---|
| Logistic Regression | 98.36% | 100%<br>97.44% | 95.65%<br>100% | 97.78 %<br>98.70% | 1 |
| Decision Tree | 91.80% | 95%<br>90% | 82%<br>97% | 88%<br>94% | 5 |
| Random Forest | 98.36% | 100%<br>97.44% | 95.65%<br>100% | 97.78%<br>98.70% | 1 |
| XGBoost | 100% | 100% | 100% | 100% | 0 |

Table 4.1 shows that XGBoost outperformed all other models by achieving perfect scores across all evaluation metrics, including 100% accuracy, precision, recall, and F1-score, with zero misclassifications. This outstanding performance reflects its advanced boosting mechanism, which incrementally builds strong learners while minimizing errors and overfitting. Its built-in regularization and fine-tuned learning strategy allowed it to capture complex patterns in the data with remarkable precision.

The Decision Tree model, on the other hand, recorded the lowest accuracy at 91.80% and had the highest number of misclassifications at five. While it demonstrated decent recall for class 1, its performance on class 0 was comparatively weaker. This suggests that the model might have overfitted the training data or lacked the complexity needed to handle certain variations in the dataset.

Random Forest performed on par with Logistic Regression, matching its 98.36% accuracy and also misclassifying just one instance. By aggregating the outputs of multiple decision trees, it managed to improve generalization and reduce overfitting. Its ability to handle non-linear patterns and diverse feature interactions contributed to its stable and consistent results.

Logistic Regression showed strong classification performance with an accuracy of 98.36%, along with high precision and recall values for both classes. With only one misclassification, it proved to be a reliable model, likely benefiting from the linear separability of the dataset. Its simplicity and efficiency allowed it to distinguish between classes effectively.

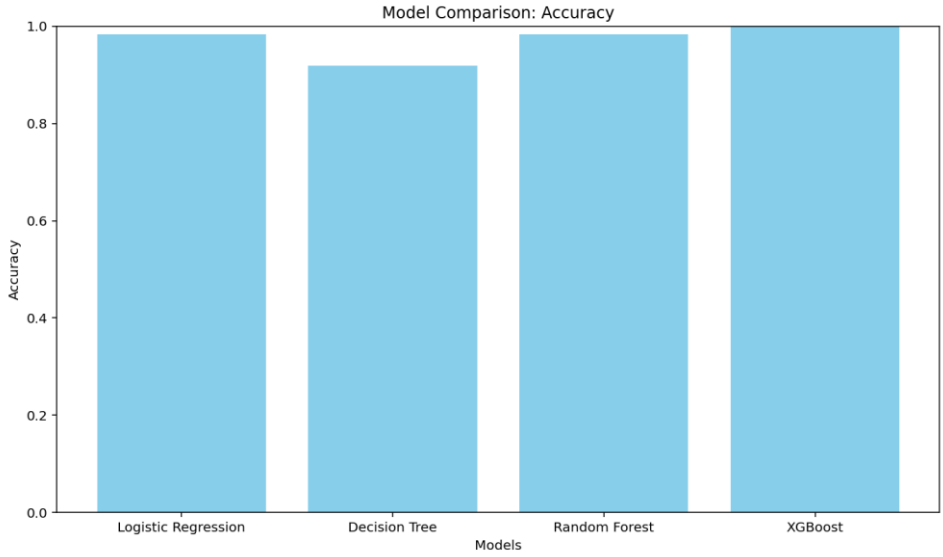



**Figure 4.10 Accuracy Comparison**

Figure 4.10 illustrates the accuracy comparison among four machine learning models: Logistic Regression, Decision Tree, Random Forest, and XGBoost. As shown, XGBoost achieved the highest accuracy, reaching a perfect score of 1.0. Logistic Regression and Random Forest also demonstrated excellent accuracy, both slightly below 1.0 at around 0.98. In contrast, the Decision Tree model showed the lowest accuracy among the group, close to 0.92, indicating reduced performance in correctly classifying instances. This visual comparison highlights the effectiveness of ensemble-based models in delivering more accurate predictions on mental health data.

## Observations:

- **XGBoost** outperformed all other models by achieving **perfect classification** across both clusters, indicating exceptional learning capability. However, such results should be cross-validated to check for overfitting.
- **Random Forest** and **Logistic Regression** both achieved **98.36% accuracy**, with only one instance misclassified. Both models showed high precision and recall, making them reliable for deployment. Random Forest provides added robustness due to ensemble learning, while Logistic Regression offers simplicity and interpretability.
- **Decision Tree**, while intuitive and easy to visualize, had a relatively lower accuracy of **91.80%**. The model struggled slightly with correctly identifying instances from Cluster 0, as seen from its lower recall (82%). It may benefit from pruning or ensemble approaches.

To gain deeper insight into the students' cognitive patterns, we analysed the average scores for five major CBT distortions, these scores were computed by taking the mean of each distortion column from the dataset yielded the following result:

**Table 5.1 Distortion Scores**

| Distortion Type | Average Score |
|---|---|
| Black & White Thinking | **1.92** |
| Personalization | 1.84 |
| Should Statements | 1.78 |
| Catastrophizing | 1.73 |
| Overgeneralization | 1.67 |

Table 5.1 depicts the most prevalent distortion as Black & White Thinking, with an average score of 1.92, indicating a frequent tendency among students to view situations in extremes (e.g., all good or all bad). By identifying the primary distortion per individual and comparing across clusters, it was found that Cluster 0 most frequently exhibited this distortion. This same cluster also had the highest average anxiety (3.01) and depression (2.77) scores, making them a high-risk group.

# CONCLUSION

The primary objective of this study was to explore mental health patterns among students and identify common cognitive distortions grounded in Cognitive Behavioural Therapy (CBT), using a blend of supervised and unsupervised machine learning approaches. This analysis aimed to generate actionable, tailored mental health interventions that address specific behavioural trends in the student population. Notably, the distortion 'Black & White Thinking' was found to be the most prevalent, with an average score of 1.92. This pattern reflects a tendency to perceive experiences in extremes, such as "If I'm not perfect, I've failed," which is often associated with heightened anxiety and depressive symptoms.

Through the application of K-Means clustering, the dataset was effectively divided into meaningful mental health groups, which allowed for focused assessment of students' psychological states. Among the classification models tested, XGBoost emerged as the most effective, achieving perfect accuracy and showing strong potential for deployment in real-world applications. This model can be utilized to automatically classify incoming student data into relevant mental health clusters, thereby enabling early identification of at-risk individuals and improving the timeliness of interventions.

One of the identified groups, Cluster 0, exhibited elevated emotional distress and high levels of cognitive distortions, indicating the need for immediate support. Students in this cluster may benefit from a range of CBT-informed interventions, including cognitive restructuring to help challenge and reframe extreme thoughts, and thought journaling to increase self-awareness and cognitive flexibility. Further, the implementation of regular mental health screenings can support early detection of concerning patterns. Complementing these strategies, the introduction of mindfulness-based workshops and structured CBT programs may foster emotional resilience. In addition, collaboration with licensed mental health professionals is crucial to ensure that students receive individualized, clinically appropriate therapeutic or medical care. The broader motivation behind this project is to develop a proactive, data-driven framework that can assist educational institutions in identifying at-risk students early and recommending targeted interventions—ultimately helping to reduce long-term psychological harm and promote overall student well-being.

# REFERENCES

## Research Papers


- Joyce H. S. You , Scotty W. C. Luk, Dilys Y. W. Chow, Xinchan Jiang, Arthur D. P. Mak, Winnie W. S. Mark , Efficacy and Cost-Effectiveness Analysis of Internet-Based Cognitive Behavioral Therapy for Obsessive-Compulsive Disorder: Randomized Controlled Trial (Published: May 5, 2022 )
- Z. Chen, R. Yang, S. Fu, and N. Zong, ―Detecting Reddit Users with Depression Using a Hybrid Neural Network, 2023.
- Khizra Khurshid1, Rabia Mushtaq2, Umara Rauf3, Nida Anwar4, Qasir Abbas1*, Sumayah Aljhani5, Zoobia Ramzan4 and Mafia Shahzadi6CBT's Impact on Procrastination, Burnout, and Anxiety in Adolescents (Published: 3 February, 2025)
- Meng Jiang, Yi Jing Yu, Qing Zhao, Jianqiang Li, Changwei Song, Hongzhi Qi, Wei Zhai, Dan Luo, Xiaoqin Wang, Guanghui Fu, Bing Xiang Yang**AI-Enhanced Cognitive Behavioral Therapy: Deep Learning and Large Language Models for Extracting Cognitive Pathways from Social Media Texts* (Published: April 17, 2024 )
- Veera Ankalu Vuyyuru, G Vamsi Krishna, S. Suma Christal Mary, S. Kayalvili, Abraheem Mohammed Sulayman Alsubayhay, Title: *A Transformer-CNN Hybrid Model for Cognitive Behavioral Therapy in Psychological Assessment and Intervention for Enhanced Diagnostic Accuracy and Treatment Efficiency* (Published: 2023)


## Website Links


- https://bmcpsychology.biomedcentral.com/articles/10.1186/s40359-025-02371-2
- https://positivepsychology.com/cbt-cognitive-behavioral-therapy-techniques-worksheets/
- https://www.cambridge.org/core/journals/european-psychiatry/article/effectiveness-of-cbt-for-children-and-adolescents-with-depression-a-systematic-review-and-metaregression-analysis/D1D82E21388A16EB077D02A24366F689
- https://neurolaunch.com/bipolar-black-and-white-thinking/
- https://zionpsychology.wordpress.com/2024/08/17/exploring-black-and-white-thinking-in-mental-health-4-examples-and-5-effective-approaches/